a

Rank

Mean Spearman, 7 targets

0.7
0.6
0.5
0.4
0.3

1
0.08M
4
0.31M
8
0.61M
16
1.23M
32
2.46M

LoRA rank (trainable weights)

b

Where LoRA goes

Mean Spearman, 7 targets

1.0
0.8
0.6
0.4
0.2
0.0

query+value, all layers
all linear layers
last 4 layers

New mutation
New position

c

Single- vs multi-task

1.0
0.8
0.6
0.4
0.2
0.0

multi-task
single-task

Fold new mut.
Fold new pos.
RAF1 new mut.
RAF1 new pos.

d

Training-set size

Mean Spearman, 7 targets

0.6
0.5
0.4
0.3

25 %
50 %
100 %

Share of training data used

e

How much of the model to train

Mean Spearman, 7 targets

1.0
0.8
0.6
0.4
0.2
0.0

head only
LoRA r=8
full fine-tuning

New mutation
New position

New mutation
New position
zero-shot ESM-2 (same colour = split)